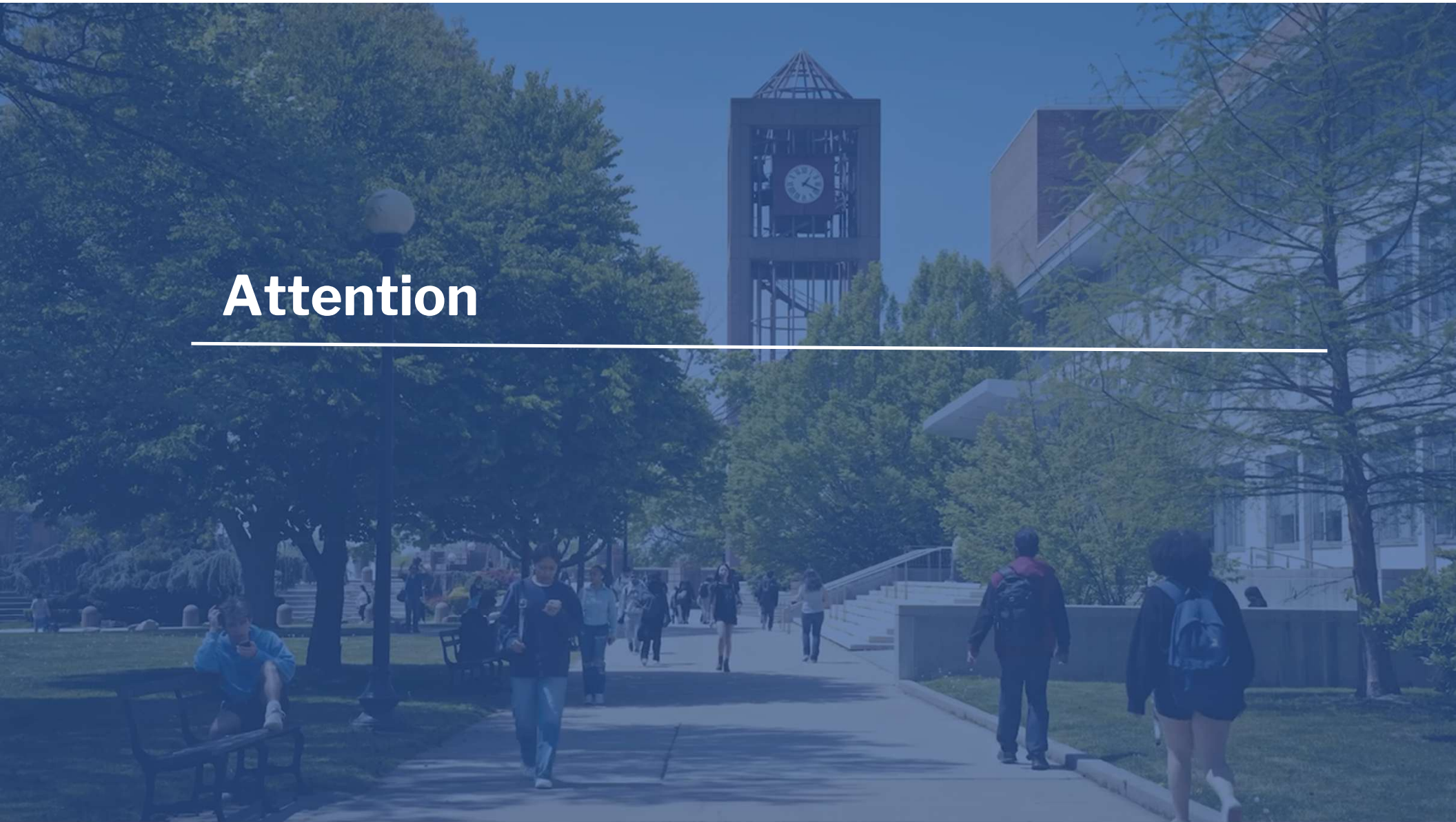


An aerial photograph of the New York City skyline, featuring numerous skyscrapers and the Hudson River in the foreground. A large blue rectangular box is centered over the image, containing white text. The sky is blue with scattered white clouds.

Introduction to Generative AI (GAI 602)

WEEK 5

Attention

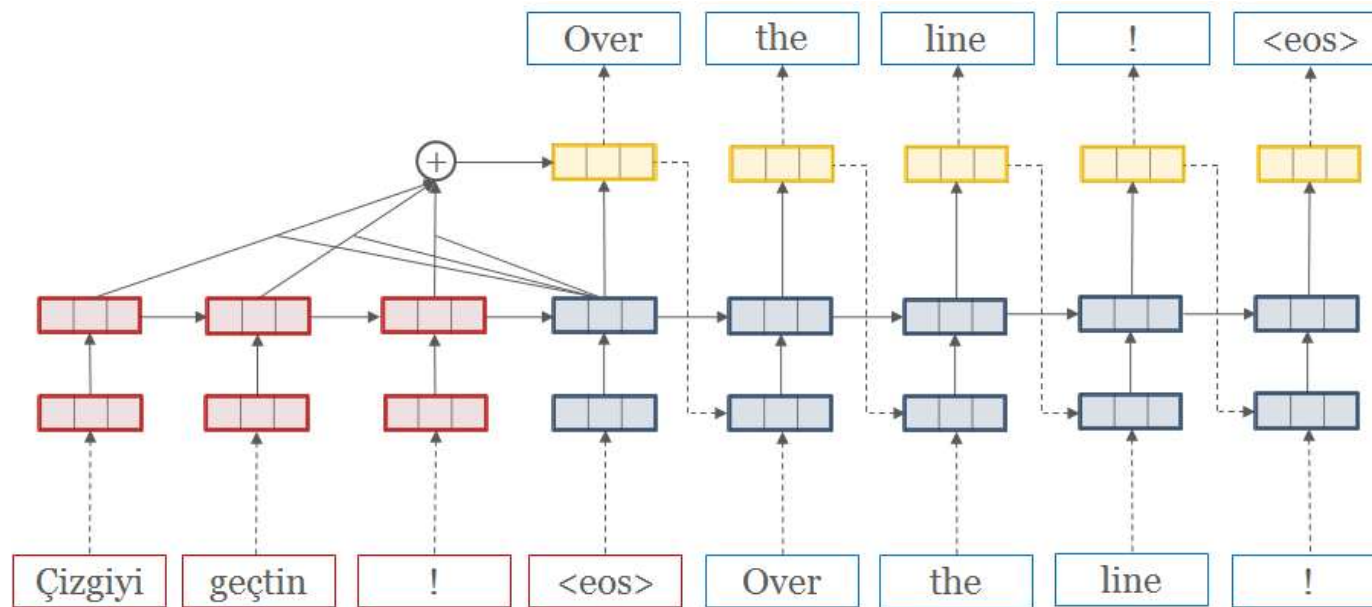


Attention

“Attention is the concentration of awareness directed at some task or phenomenon while mostly excluding others”

William James, “Principles of Psychology” (1890)

Traditional Attention



Source: opennmt.net

Attention is all you need – Vaswani et al 2017

- New concept of how to make use of attention
- Completely removed the reliance on recurrence to process sequences.

Attention Is All You Need

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Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature. We show that the Transformer generalizes well to other tasks by applying it successfully to English constituency parsing both with large and limited training data.

*Equal contribution. Listing order is random. Jakob proposed replacing RNNs with self-attention and started the effort to evaluate this idea. Ashish, with Illia, designed and implemented the first Transformer models and has been crucially involved in every aspect of this work. Noam proposed scaled dot-product attention, multi-head attention and the parameter-free position representation and became the other person involved in nearly every detail. Niki designed, implemented, tuned and evaluated countless model variants in our original codebase and tensor2tensor. Llion also experimented with novel model variants, was responsible for our initial codebase, and efficient inference and visualizations. Lukasz and Aidan spent countless long days designing various parts of and implementing tensor2tensor, replacing our earlier codebase, greatly improving results and massively accelerating our research.

[†]Work performed while at Google Brain.

[‡]Work performed while at Google Research.

31st Conference on Neural Information Processing Systems (NIPS 2017), Long Beach, CA, USA.

Source: Nvidia

Self-attention

Self-Attention: What Does It Actually Mean?

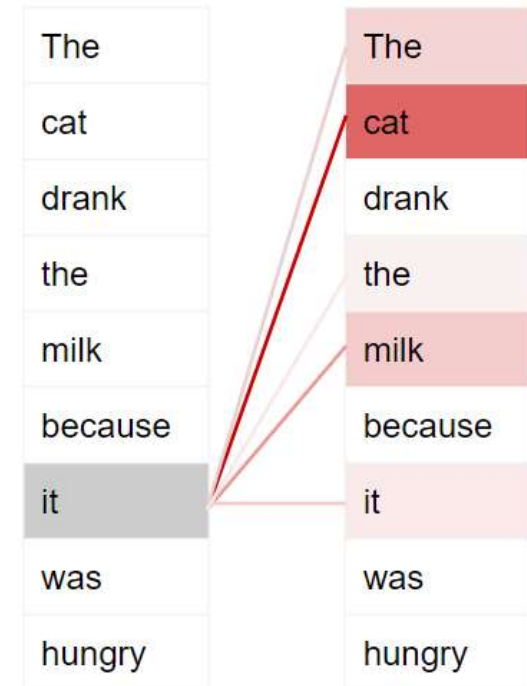
Self-attention is a mechanism that allows a model to dynamically focus on the most relevant parts of a sequence when computing representations for each token.

Key Concept

Every token in the sequence can **attend** to every other token, including itself, to understand its relationship and importance in the context of the entire sequence.

For example

In the sentence: "The cat drank the milk because it was hungry". Self-attention helps the model understand that "it" refers to the "cat".



Source: Nvidia

How Self-Attention Mechanisms Work

The implementation of self-attention can vary, but the core concepts are:

- **Compare:** Each token is compared to every other token in the sequence to compute relevance scores
- **Weight:** Assign weights to other tokens based on their relevance to the current token
- **Aggregate:** Combine information from all tokens, weighted by their relevance, to compute a new representation for the current token.

Global Context

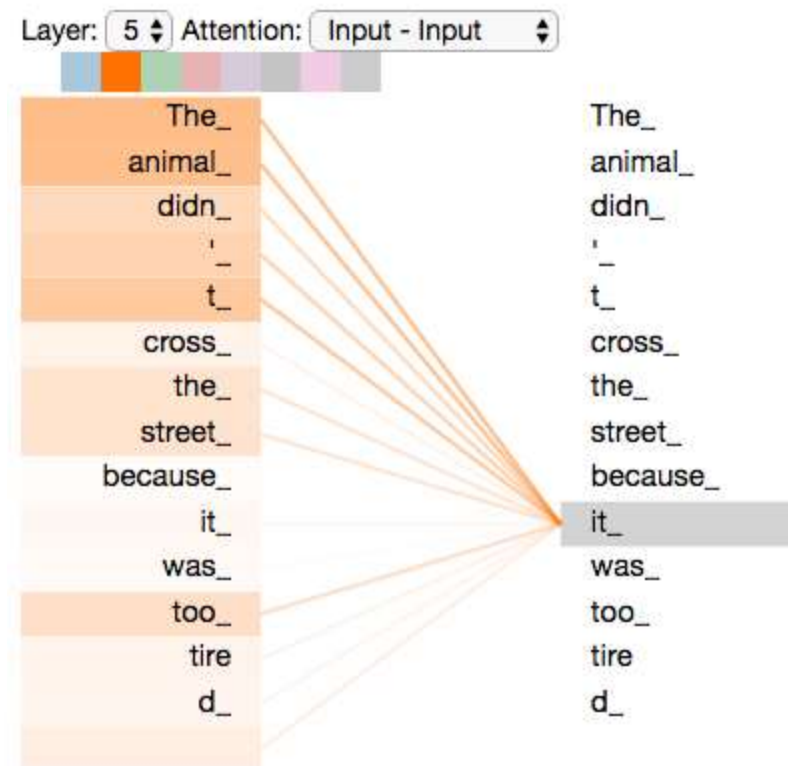
Captures relationships between tokens across the entire sequence, not just local neighbors.

Dynamic Focus

The model decides what to focus on for each token, rather than relying on fixed patterns (e.g., sliding windows in convolution).

Flexibility

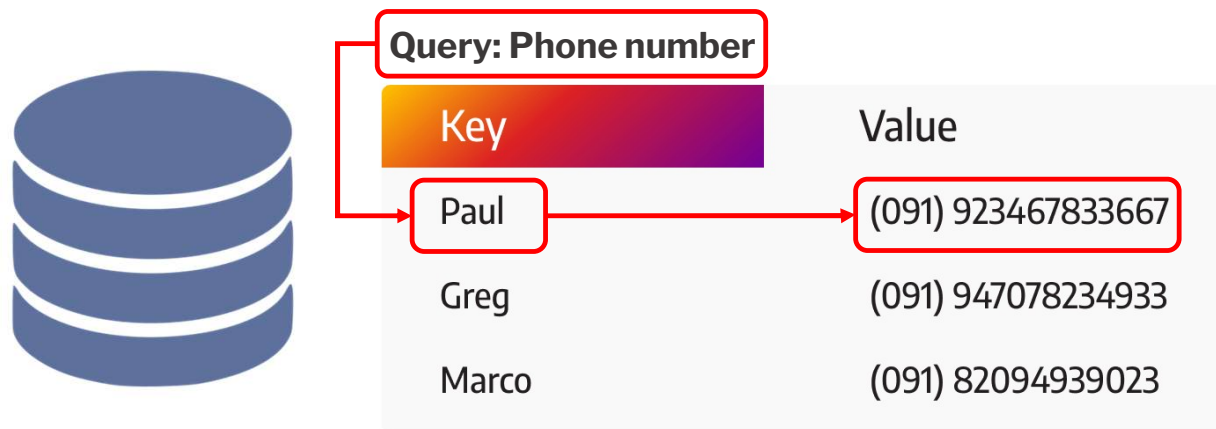
Works for variable-length sequences and tasks requiring both short- and long-range dependencies.



Source: Nvidia

Database Analogy: Query-Key Value

Example: Looking up a phone number in a database



In databases **Query=Key** always. We want an **exact** match.

But what if we want to have fuzzy matches?

Source: Memgraph

Query Key and Value

In the Attention is all you need paper, an algorithm, based on Queries, Keys, and Values is presented to handle the attention calculations:

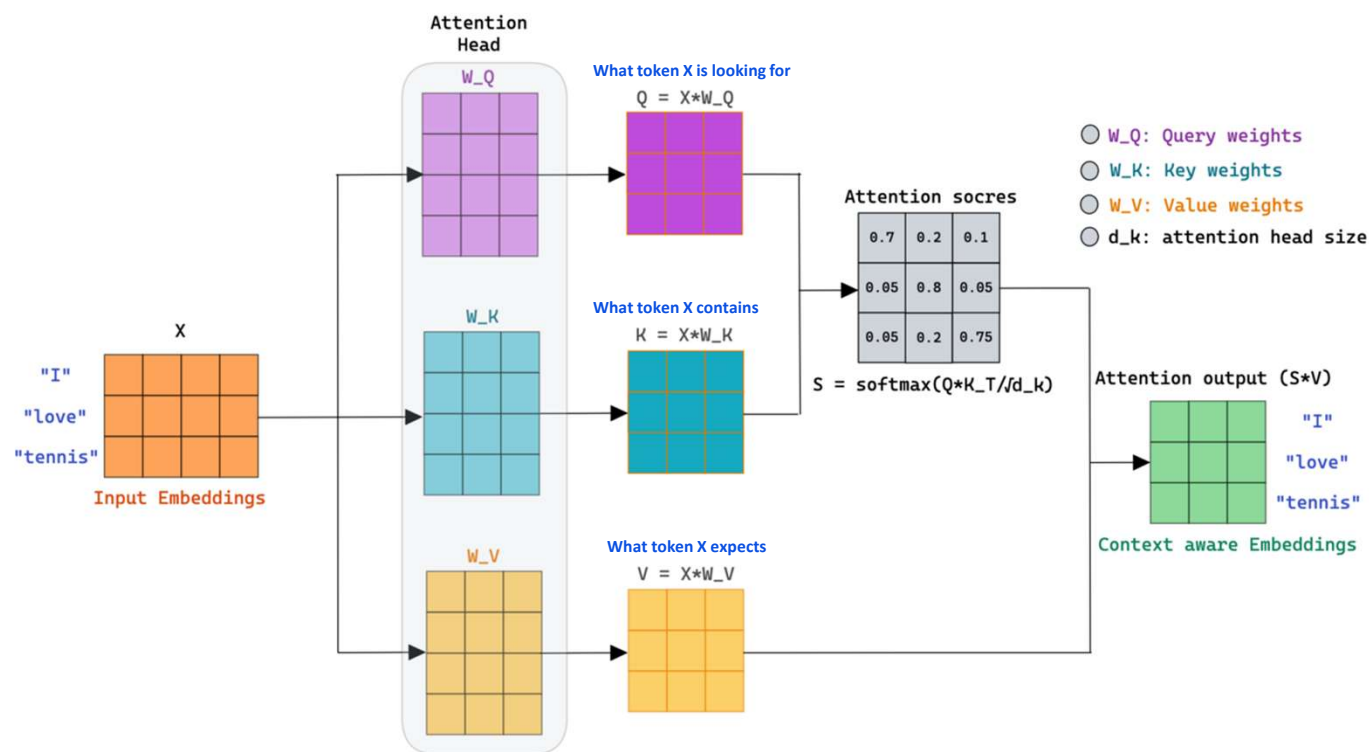
- **Query:** What it wants to know
- **Key:** What information it can provide
- **Value:** What (embedding) information it actually passes along

Process

- **Step 1: Create Q, K, V**
Each input token (e.g. embedding) is linearly transformed into three vectors: Query (Q), Key (K), and Value (V).
- **Step 2: Compute Attention Scores:**
Compare the Query of a token with the Keys of all tokens in the sequence to measure relevance. Use the dot product to capture similarity.
- **Step 3: Normalize Scores (SoftMax):**
Apply SoftMax to the attention scores to ensure they sum to 1, producing attention weights.
- **Step 4: Aggregate Values:**
Multiply the attention weights by the corresponding Values and sum them to produce the output representation for each token.

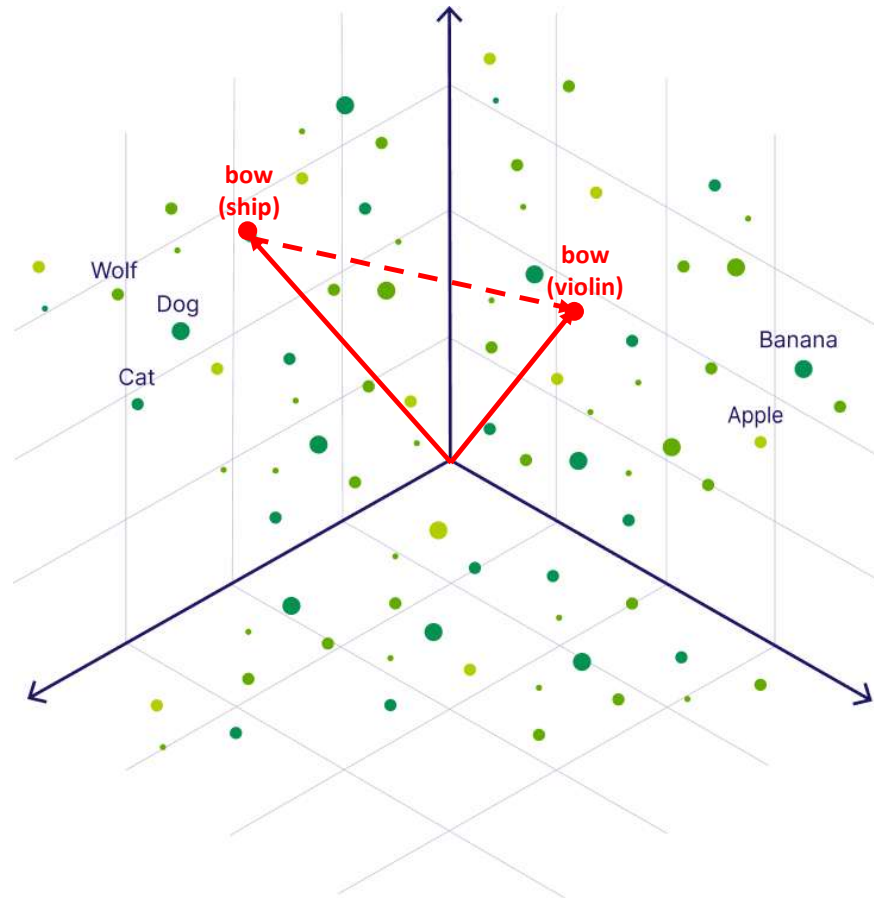
Source: Nvidia

Attention



Source: Nvidia

Attention



Source: Nvidia

Visualization

TRANSFORMER EXPLAINER

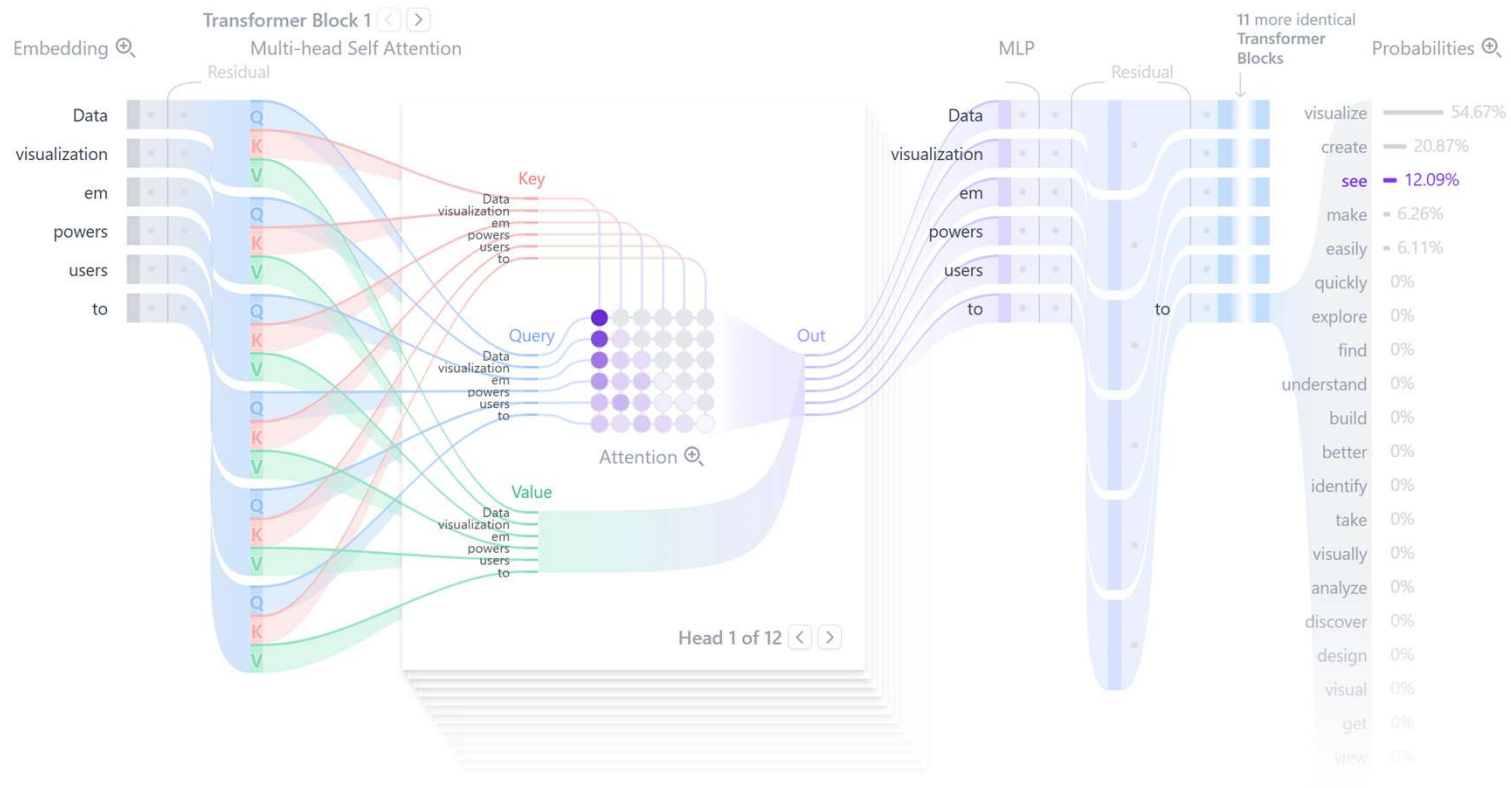
Examples ▾

Data visualization empowers users to see

Generate

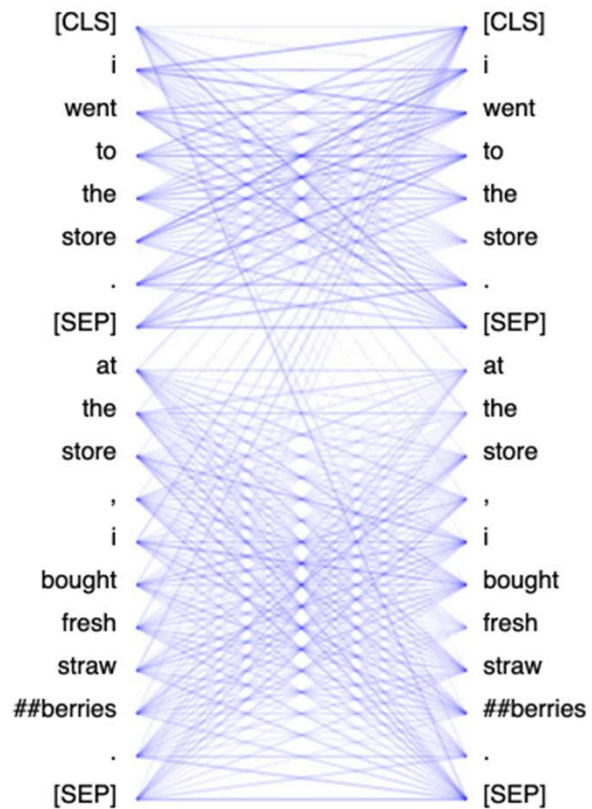
Temperature 0.8

Sampling ☒ Top-k ☐ Top-p k=5



Source: <https://poloclub.github.io/transformer-explainer/>

Visualization



Source: Comet

Attention

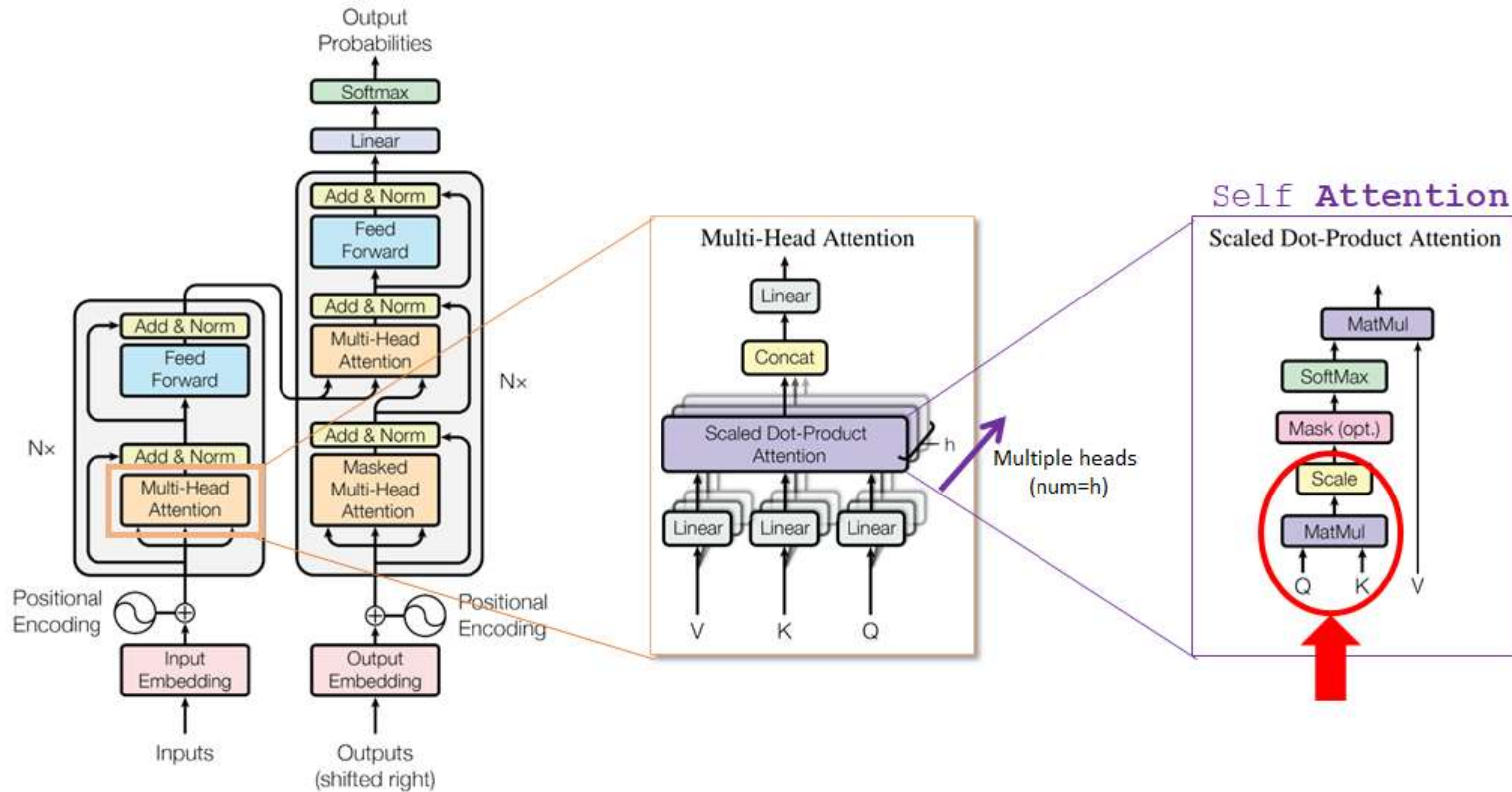


Figure 1: The Transformer - model architecture.

Source: Nvidia

Benefits of Attention

- **Capturing Long-Range Context and Dependencies**

The use of the query, key, and value concept allows transformers to capture context and dependencies between words. This makes them highly effective in a wide range of applications, from NLP to image recognition.

- **Improving Accuracy**

Because attention dynamically adapts to each input, models learn context-specific representations rather than fixed encodings, leading to higher accuracy across domains.

- **Scaling to Large Datasets**

The Query–Key–Value (QKV) mechanism scales naturally to massive datasets, deep architectures, and large model sizes. This is a key enabler of foundation models.

- **Selective Focus on Relevant Information**

By weighting inputs via attention scores, Transformers prioritize the most informative parts of the input. This reduces noise and improves performance on tasks like translation, summarization, and reasoning.

- **Parallelization and Training Efficiency**

Unlike sequential models, attention enables full parallel computation over tokens. This dramatically accelerates training and inference on modern hardware (GPUs/TPUs)..

- **Multi-Perspective Understanding via Multi-Head Attention**

Multiple attention heads allow the model to attend to different relationships simultaneously (syntax, semantics, positional structure, etc.), improving robustness and generalization.

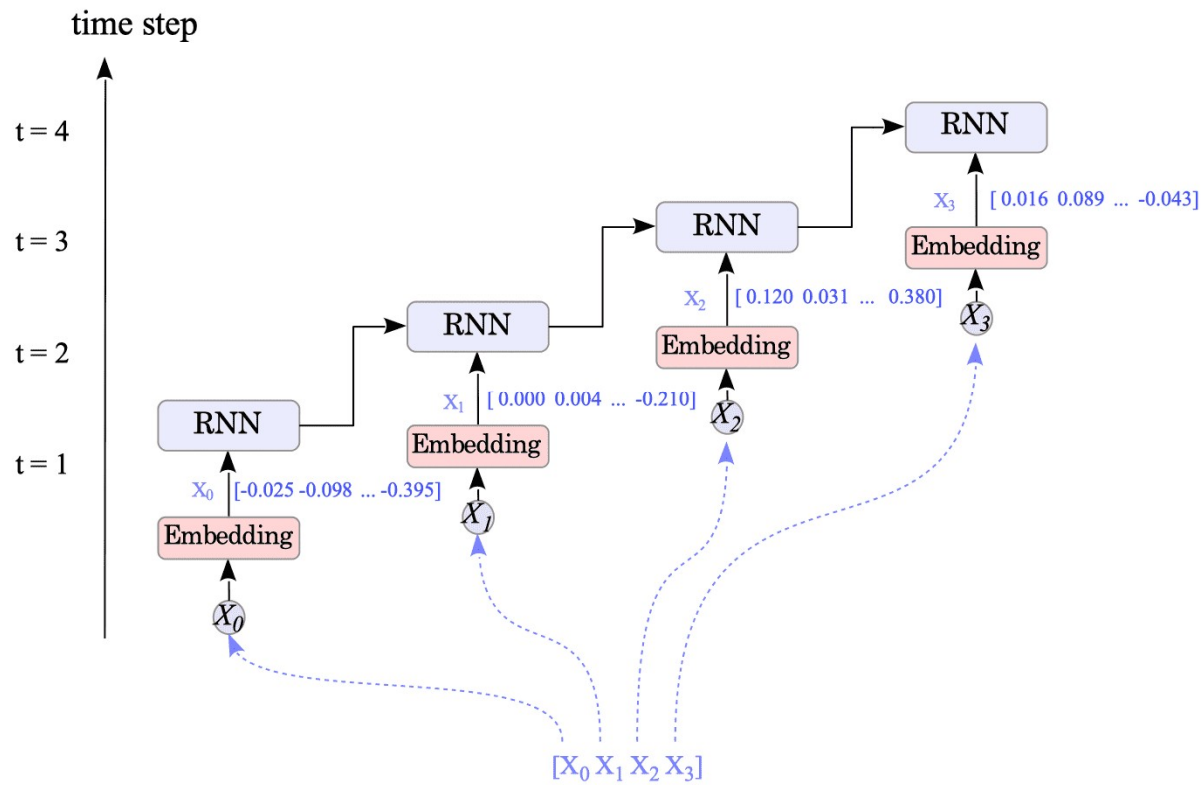
Source: IBM

Positional Encoding



RNN Positional Encoding

Positions of tokens is implicit (it is part of the hidden state).



Source: IBM

Position matters in Language

dog bites man vs man bites dog

**“When a dog bites a man that is not news,
but when a man bites a dog that is news”**

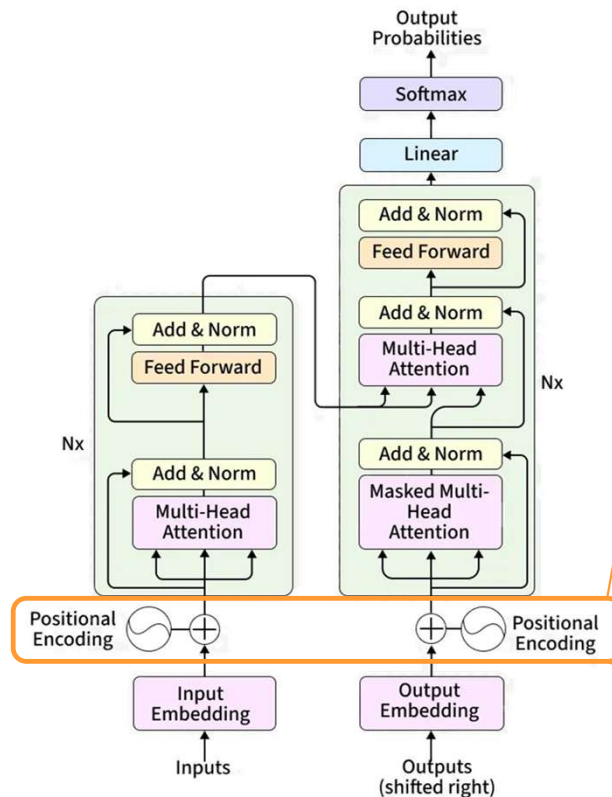
US newspaper magnate Charles Anderson Dana



Source: IBM

Positional Encoding

Transformers don't process tokens sequentially, so there is no position information. The sinusoidal position encoding formulae, defined by the authors of the original transformers paper.



For even positions:

$$PE_{pos,2i} = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

For odd positions:

$$PE_{pos,2i+1} = \cos\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

k : The position of the word in the sentence (e.g. 0 for the 1st word, 1 for the 2nd, and so on.)

i : The dimension index of the embedding vector. maps to column index. $2i$ will indicate an even position and $2i+1$ will indicate an odd position

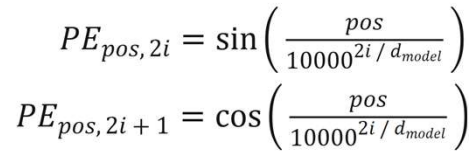
d_{model} : The predefined dimensionality of the token embeddings (for example, 512)

n : user-defined scaler value (for example, 10000)

PE : position function for mapping position k in the input sequence to get the positional mapping

Source: IBM

Let's look at the encoding pattern



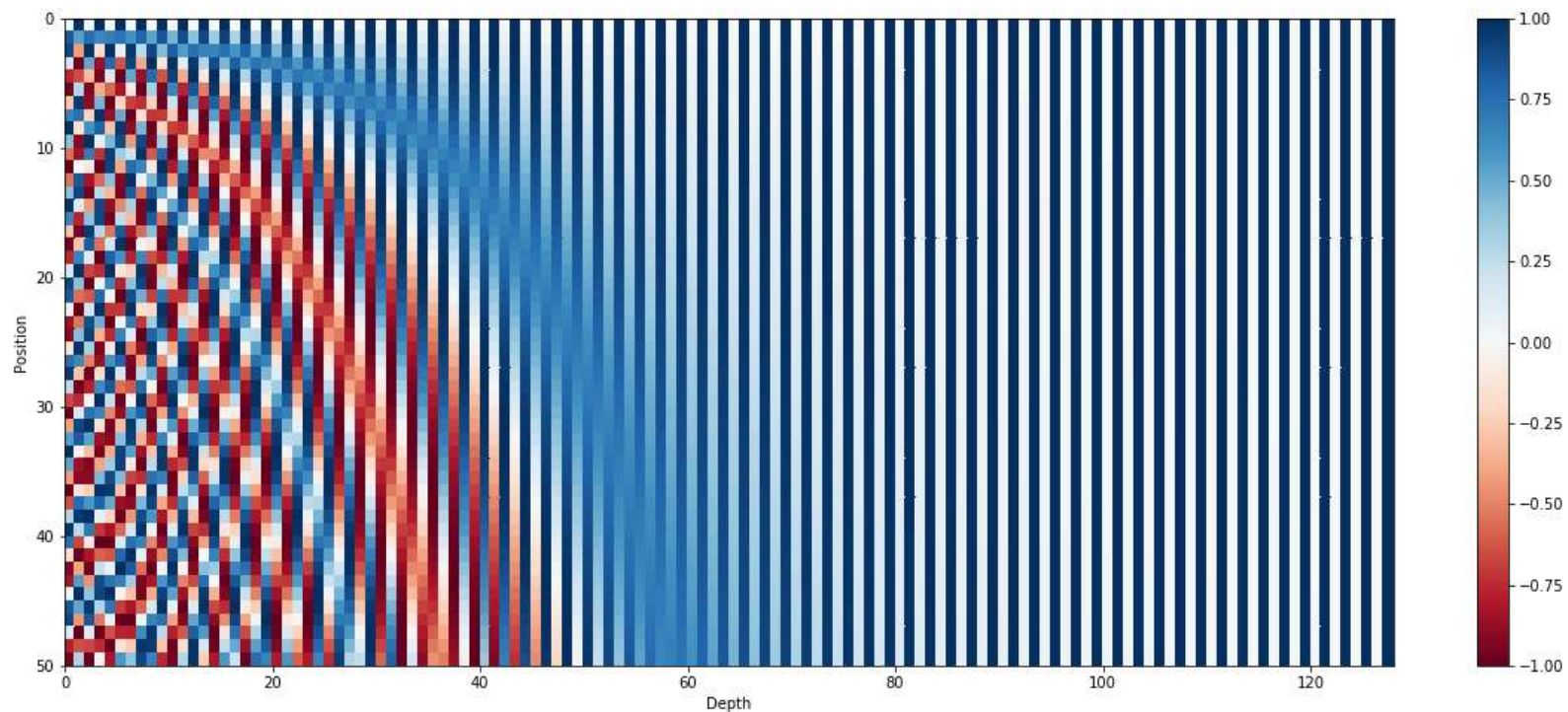
Let's consider the pattern that the numbers 0-127 make in binary encoding:

Similar patterns surface for other bases.

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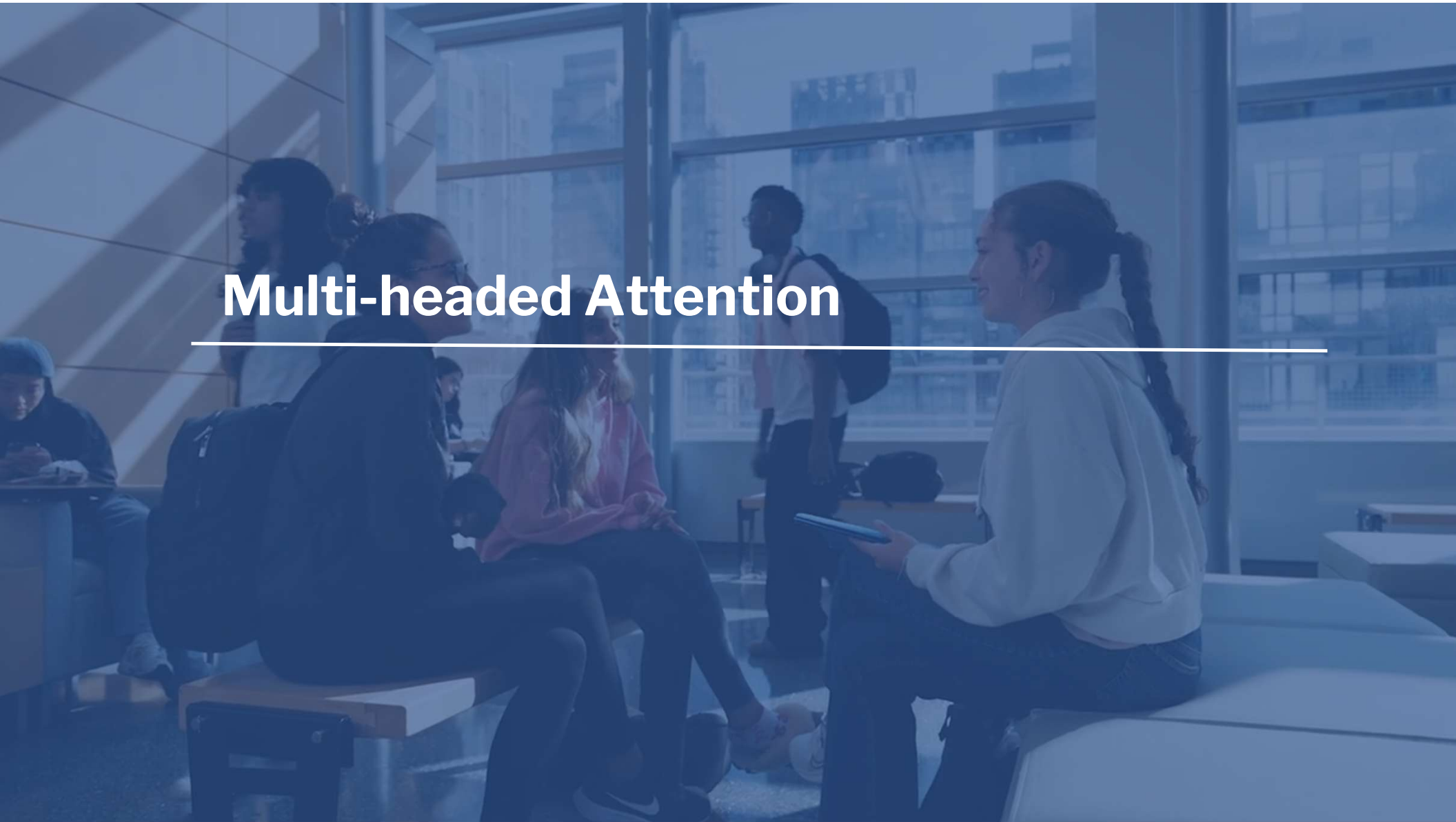
Visualizing Transformer Position Encoding

Positional encoding allows the transformer to learn position and relative distance between tokens, based on the encoding pattern.



Source: ATFSAD: Enhancing Long Sequence Time-Series Forecasting on Air Temperature Prediction, Bin Yang, Tinghua Ma, and Xuejian Huang

Multi-headed Attention



Self-Attention – Multiheaded Attention

In the paper, the authors also implemented “multi-headed attention” to address the limitations of single-headed attention, which could only focus on one type of relationship or dependency at a time.

Parallel Attention Mechanisms

Multi-headed attention allows the model to focus on different parts of the input sequence simultaneously, capturing diverse relationships (e.g., word-to-word, phrase-to-phrase).

Improved Representations:

Each “head” learns unique attention patterns, enabling the model to extract more nuanced information, such as syntactic and semantic dependencies.

Scalability and Flexibility:

By splitting the computation into smaller attention heads, the model efficiently handles large inputs and produces richer, context-aware embeddings.

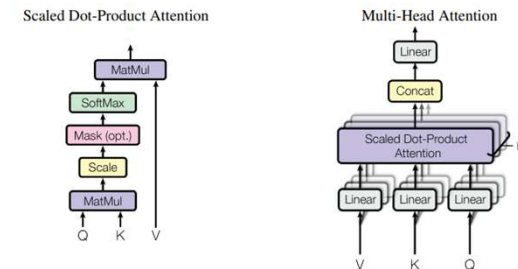
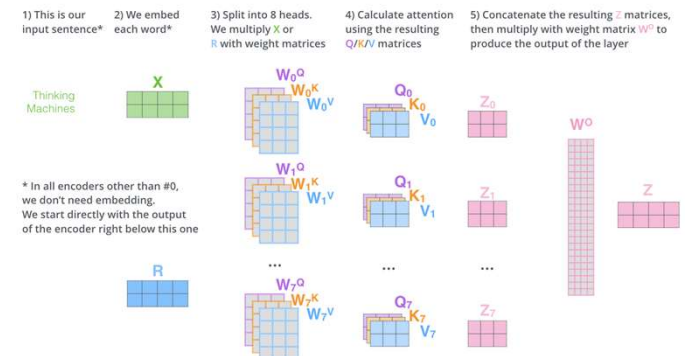


Figure 2: (left) Scaled Dot-Product Attention. (right) Multi-Head Attention consists of several attention layers running in parallel.



Source: Nvidia

Variations of Multiheaded Attention

GQA (Grouped-Query Attention)

Reduces the computational cost by grouping multiple queries together and computing attention collectively rather than individually

Key Advantage: Drastically lowers memory and compute requirements, especially for large-scale models like GPT-3

Multi-Query Attention:

Instead of having separate key-value pairs for each query, all queries share the same set of key-value pairs

Key Advantage: Reduces the required memory for storing multiple key-value pairs without impacting model performance

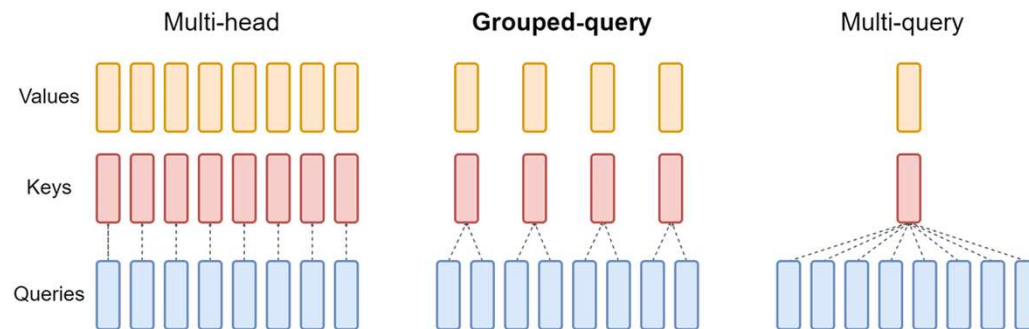


Figure 2: Overview of grouped-query method. Multi-head attention has H query, key, and value heads. Multi-query attention shares single key and value heads across all query heads. Grouped-query attention instead shares single key and value heads for each *group* of query heads, interpolating between multi-head and multi-query attention.

Source: Nvidia

An aerial photograph of the Manhattan skyline, featuring the Manhattan Bridge in the foreground and the dense cityscape in the background. The bridge's stone towers and suspension cables are prominent. The water of the harbor is visible in the lower foreground. The sky is clear and blue.

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